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Review

Biological and mechanical measures for runoff and soil erosion control in India and beyond

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Abstract

Soil erosion poses a significant threat to agricultural sustainability, particularly in India, where approximately 83.0 million hectares of land are affected, with an average soil loss of 16.23 t ha^{-1} annually. This review systematically analyzes control practices for mitigating soil erosion in agricultural lands, focusing on their effectiveness in reducing runoff and soil loss. The findings highlight that biological methods, such as mulching and intercropping, are widely adopted, with mulching reducing soil loss and runoff by 5–18% and 3–15%, respectively. Tillage operations, particularly conservation tillage, have shown promise, reducing soil loss by 14–18% and improving infiltration rates. Mechanical methods, including contour bunds and terracing, further complement these efforts, achieving reductions in soil loss by 44–52% and runoff by 36–46%. The review underscores the importance of integrating biological and mechanical approaches for optimal erosion control, as well as the need for policy support and farmer education to promote widespread adoption. By addressing soil erosion through these practices, agricultural productivity, soil fertility, and environmental health can be significantly enhanced, contributing to global food security and sustainable land management goals.

Article Highlights

- Mulching, intercropping & no-till farming cut soil loss – proven methods to protect farmland sustainably.
- Mix plants + structures for best results – combined approaches boost long-term soil protection.
- Farmers + policies must team up – scaling these practices ensures food security for the future.

Keywords Agricultural land · Biological methods · Mechanical methods · Runoff · Soil erosion

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1 Introduction

Land is a vital natural resource that supports all forms of life. In 2015, the United Nations proposed the Sustainable Development Goals (SDGs), which aim to create a more sustainable future for all. The SDGs related to food (SDG 2), health (SDG 3), water (SDG 6), and climate (SDG 13) amplify pressure on land through competing demands. Expanding agriculture to meet rising food needs risks deforestation and soil erosion, while health and nutrition goals drive land use for diverse crops and pollution-sensitive ecosystems. Water security requires land for irrigation and infrastructure, intensifying scarcity and degradation. Climate action prioritizes land for carbon sequestration (e.g., forests, bioenergy), conflicting with agriculture. Anticipating a global population surpassing nine billion by 2050, a 60% surge in the need for food, feed, and fiber is foreseen [1, 2, 87, 89]. To meet these needs, it is necessary to use land for agriculture and development, but this can also lead to land degradation issues. To address these issues, it is essential to halt land degradation and implement sustainable land management practices [3]. SDG 15.3 focuses specifically on achieving land degradation neutrality. Given that soil is an integral component of land, degradation of land has repercussions on soil systems, which are intricately linked to other ecosystems, including water. As such, soil science is an important land-related field that is linked to several SDGs and focuses on soil erosion as a major concern [2, 4].

Soil erosion is a widespread problem and a major concern in many countries. In India, it is a critical issue with 5.3 billion tons of soil lost annually, resulting in an economic loss of ₹89,000 crore due to reduced productivity and siltation. Similar impacts are observed globally—Ethiopia loses 1.5 billion tons/year, China's Loess Plateau experiences erosion rates over 5,000 tons/km^2/year, and the U.S. loses 2.7 billion tons/year, costing $44 billion Moreover, it refers to the destruction of soil caused by both natural forces (such as water and wind) and human activities [5]. Erosion can occur naturally or as an accelerated process due to human activities [6]. Natural erosion happens over millions of years and helps form new soil, while accelerated erosion is primarily caused by practices like deforestation, overgrazing, shifting cultivation and poor farming techniques, resulting in excessive soil loss. Soil erosion rates in shifting cultivation plots reached 15–30 Mg ha^{-1} yr^{-1} on slopes > 20%, compared to < 5 Mg ha^{-1} yr^{-1} in mature forests [90, 91]. This erosion has detrimental effects on agricultural production, water quality, and the health of ecosystems in both aquatic and land environments. Soil erosion factors include ground cover, climate, rainfall erosivity soil erodibility, and terrain [7, 8, 85].

Soil is essential for agriculture production, but soil erosion in agricultural areas poses a threat to the sustainability of farming. Accelerated soil erosion has negative environmental and economic effects, impacting on-site and off-site productivity due to factors like declining soil quality and short- and long-term effects [9]. In agriculture, improper practices such as intensive tillage contribute to soil detachment during rain, especially in continuous heavy rainfall. Studies have explored the impact of crop rotation on soil erosion rates [10–12]. One of the consequences of soil erosion is the loss of topsoil and a decrease in soil fertility, making the land less suitable for agriculture and decreasing crop yields. Additionally, the fine particles of the eroded sediment that eventually reach surface water bodies can also lead to problems such as high sedimentation and flooding. If the eroded material contains pesticides or fertilizers, it can also degrade downstream water quality and harm aquatic organisms [7].

India produces a majority of the world's food, primarily for local consumption. Additionally, the Indian continent is known for its cultivation of important crops such as rice, wheat, maize and sugarcane [13, 14]. However, despite being a major source of income and livelihood for millions in India, but certain farming practices can have severe environmental consequences. Some of the most damaging activities include excessive tilling, which breaks down soil structure and makes it vulnerable to erosion; mono-cropping, which depletes soil nutrients over time and increases erosion risk; and overgrazing, which strips away protective vegetation cover. Additionally, slash-and-burn cultivation and improper irrigation methods further degrade land by accelerating topsoil loss and reducing water retention capacity. These unsustainable practices contribute highly to soil erosion. Globally, approximately 24 billion tonnes of fertile soil are lost annually due to water erosion. India is among the most severely affected countries, with an estimated 120.70 million hectares (Mha) impacted by land degradation. The primary contributor is water erosion, affecting around 83 Mha (68.4%), followed by wind erosion (12.4 Mha). Chemical degradation, including soil salinity (6.74 Mha) and acidity (17.94 Mha), as well as physical degradation from mining and waterlogging (1.07 Mha), further reduce land productivity. Combined, water and wind erosion account for 95.4 Mha of degraded land, with an average soil loss rate of 16.23 t ha^{-1} affected [7, 15, 85]. Several regions, including North Himalaya, Nilgiri, and coastal, have been identified as "hot-spots" with erosion rates higher than permissible limit [16]. India is among the most affected countries, with one third of their total geographical area as shown in Table 1. The highest soil erosion rate is found in agricultural land due to inappropriate practices such as farming on steep slopes and cultivating without protective measures [17].

Table 1 Worldwide soil erosion status

Country	Geographical area (M ha)	The area under land degradation (M ha)	Average soil loss (t ha^{-1})
Afghanistan	64.18	–	9.12
Albania	2.87	–	12.31
Argentina	278.24	90.2	2.21
Australia	769.18	199.4	0.75
Brazil	846.71	188.2	4.32
Burundi	2.70		37.12
Canada	992.30	198.5	1.26
China	937.30	219.4	6.4
Comoros	0.15	–	35.60
Ethiopia	112.70	–	12.60
Fiji	1.83	–	12.73
Haiti	2.67	–	53.60
India	328.72	120.7	16.23
Indonesia	186.39	102.9	6.11
Iran	162.18	2.9	2.14
Japan	37.02	13.1	4.43
Nepal	14.69	5.5	2.93
New Zealand	26.74	14.7	2.94
Philippines	28.91	–	15.6
Portugal	9.08	–	1.19
Russia	1697.35	280.2	0.97
South Africa	122.00	35.2	3.37
Sri Lanka	6.57	2.1	3.78
Switzerland	4.13		5.66
United States of America	946.59	198.4	2.18

To address the negative effects of soil erosion on agricultural land, researchers have proposed strategies and control practices to reduce the soil erosion rate [16]. Controlling soil erosion involves implementing innovative land management techniques and methods. A lot of research is conducted each year on the management of soil erosion in worldwide, focusing on the effects of various management practices such as tillage operations, mulching, cover crops, and intercropping on runoff and erosion [6, 7]. However, the findings of these studies are sometimes conflicting, which may be due to differences in study design, flaws, or chance variation. Thus, a systematic review is needed to identify the research done and to identify the most reliable results that can be applied.

A systematic review is a methodical examination of research related to a specific question, in which explicit and systematic procedures are employed to identify, select, and critically evaluate relevant studies, and to collect and analyze data from the included studies. Statistical analysis may or may not be used to summarize the results. This systematic review aims to evaluate the efficacy of biological (mulching, intercropping) and mechanical (terracing, contour bunds) erosion control practices across India's agro-climatic zones, quantifying their impacts on soil loss and runoff reduction. It analyzes regional suitability, focusing on smallholder systems in high-erosion hotspots. The review advances beyond prior work by combining traditional and emerging techniques while aligning with SDG 15.3 targets. It provides context-specific guidance to enhance sustainable land management for policymakers and farmers.

2 Systematic reviews and meta-analyses

This section covered the methodology employed for retrieving articles on soil erosion control practices in agricultural land in India. The systematic literature review was conducted using a methodology known as PRISMA, which encompassed a number of components, such as identifying resources, establishing eligibility and exclusion criteria, outlining

the systematic review process, and conducting data abstraction and analysis. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines was followed to ensure the completeness and transparency of the review process (93).

2.1 Information sources and search

The following electronic databases were searched: Web of Science and Google Scholar. In addition, the reference lists of relevant articles were hand-searched to identify additional studies. The search included all studies published in the English language from the year 2015 to 2023. Keywords and search terms included "soil erosion control", "agricultural land", "India", "world", "land management practices", "conservation practices", "crop management", "vegetative measures", "engineering measures", "erosion control structures", "farming practices", "land use change", and "land degradation". Boolean operators were used to combine keywords and search terms.

2.2 Eligibility and exclusion criteria

The eligibility criteria for the studies was included in the review are as follows: (1) studies that investigate the effectiveness of soil erosion control practices on agricultural land in India; (2) studies that report original data; (3) studies that use quantitative or qualitative methods; (4) studies that are published in English language from 2015 to 2023. The exclusion criteria are: (1) studies that are not related to soil erosion control practices; (2) studies that focus on non-agricultural land; (3) studies that do not report original data; (4) studies that are not published in English language.

3 Thematic analyses

The systematic review produced two main themes and several sub-themes related to soil erosion control practices. The main themes are agronomic practices and mechanical or engineering practices. The sub-themes are contour terracing, bunds, contour farming, geo-textile, organic matter, mulching, cover crop, intercropping, tillage operation, and cultivation of grass.

3.1 Mulching

Mulching is a practice of covering the soil surface with a layer of organic or inorganic material, such as straw, leaves, plastic, or rocks, to conserve soil moisture, reduce soil erosion, and suppress weed growth. Mulching can also impact runoff and soil loss by reducing the kinetic energy of raindrops, promoting water infiltration, and enhancing soil structure and stability. Mulching also improves soil quality and fertility, which can indirectly affect runoff and soil loss as shown in Table 2. A study examined the impact of mulching on soil properties and water dynamics in a degraded soil of India. It was found that mulching with crop residues increased soil organic matter content, total porosity, water holding capacity, and hydraulic conductivity, and decreased bulk density. As a result, mulching increased infiltration and decreased runoff and soil loss by 36% compared to no mulching [18]. In a separate study, researchers evaluated the effect of pearl millet stubble mulching on soil erosion within a green gram/cowpea-mustard cropping system. The study found that mulching reduced soil loss by 59.6% and increased infiltration and water retention in soil [19]. The authors concluded that pearl millet stubble mulching could enhance soil fertility, crop productivity, and environmental sustainability in smallholder farming systems. A study conducted in a semi-arid region in Iran, testing three treatments: removing plant residues, returning residues to the soil surface after harvesting, and burning plant residue [20]. Results revealed a 96.5% reduction in soil loss with the returning residue treatment compared to the control, while the burning residue treatment showed a 192% increase erosion and soil loss. Additionally, the returning residue treatment exhibited the lowest runoff volume among the three treatments.

Studies have examined the efficacy of mulching techniques in controlling soil erosion. In a study it was found that combining reduced tillage practices with straw returning as mulch significantly increased soil water content in the 0–10 cm soil depth, leading to higher infiltration rates and lower runoff rates [21]. Straw mulching can decrease runoff depths and sediment yields by 81% and 97%, respectively [22, 85]. Results demonstrated that rice straw mulch applied at 5 t ha^{-1} achieved the lowest average soil detachment rate across all treatments [23]. Similarly, in a study it was found

Table 2 Impact of mulching on runoff and soil loss/erosion

Mulching material	Soil type	Quantity (Mg ha^{-1})	Crop/cropping system	Reduction in runoff (%)	Reduction in soil loss (%)	Reference
Banana leaves mulch + compost	Loam	4.0	Maize + bean	99	100	[26]
Guinea grass mulch + pigeon-pea hedges + animal manure	Sandy loam	4.0		99	100	
Guinea grass mulch + pigeon-pea hedges + green manure	Silt–clay loam	4.0		95	100	
Rice straw mulch	Sandy loam	6.0	Maize	62.7	87.7	[27]
Fresh mustard biomass	Loamy	3.0	Corn	–	53.9	[28]
Rice straw mulch	Clay (Ultisols)	2.0	Peanut	55.0	91.4	[29]
Swine manure				12.0	75.9	
Plastic mulch	Silt loam		Maize + cowpea	36.0	41	[18]
Maize residue mulch	Sandy clayey	1.5–4.5	Maize	10–50	50–90	[30]
Recommended dose of fertilizers + gypsum + crop residue mulch	Silty clay loam	1.5–3.0	Soybean	25	29	[31]
Pearl millet stubble mulch	Sandy loam soil	2.0	Green gram/cowpea-mustard, pearl millet + pigeon pea and cowpea + castor	59.6	69.2	[19]
Basal fertilizers + wheat straw mulch + tied ridges	Sandy clay loam	15.0	Maize	87.2	81.4	[32]
Straw mulch	Clay	–	Rapeseed + peanut	81	91	[22]

that straw mulch helped to reduce surface runoff [24]. Another study investigated wheat straw mulch effects on soil loss and runoff, revealing that optimal application rates—dependent on rainfall intensity, soil type, and slope gradient—could significantly reduce erosion [25].

3.2 Intercropping

Intercropping is a traditional and sustainable agricultural practice that involves the cultivation of two or more crops in a definite row proportion on the same field simultaneously or sequentially. It is a sustainable approach to farming that has been shown to have a positive impact on soil health [86, 88], water conservation, and pest management. Research has demonstrated that intercropping can be an effective method for reducing runoff and soil erosion in agricultural areas, both in India and globally. A study in northern India found that intercropping maize and different crop combinations reduced soil loss by 61% and increased infiltration by 23% compared to sole cropping. [15]. A study in India found that intercropping of maize with legumes reduced soil erosion and runoff by up to 51% and 41%, respectively [33]. Similarly, a study in India found that intercropping maize and cowpea reduced soil loss by 36% compared to sole cropping. [18]. These results demonstrate the potential for intercropping to be an effective method for reducing runoff and soil loss in agricultural systems both in India and globally. In Brazil, intercropping of corn and brachiaria grass reduced soil erosion and runoff by up to 66% and 78%, respectively [34]. Another study in China showed that intercropping of corn and soybean reduced runoff and soil loss compared to monoculture crops [35].

To tackle the pressing challenge of soil erosion in the fragile terrains of the Himalayan Mountains where steep slopes render agricultural lands highly susceptible a field experiment was carried out focusing on intercropping as a potential solution [33]. The research involved cultivating maize during the rainy season, followed by rainfed wheat in the winter season (November–April), and incorporating okra and cowpea as intercropping crops. Results revealed a 26% reduction in runoff and a 43% decrease in soil loss when one row of cowpea or okra was planted between two rows of maize. Intercropping has also been shown to improve soil structure, increase soil organic matter content, and reduce nutrient loss, making it a sustainable approach to agriculture.

3.3 Organic matter

Decomposition of organic inputs and manure can play a significant role in reducing soil erosion by improving soil quality and structure. An Indian investigation revealed that incorporating organic inputs into soil management practices significantly enhanced soil vitality and curbed erosion. This improvement stemmed from increased soil moisture retention, elevated levels of organic matter, and better structural integrity of the soil matrix [36]. The researchers concluded that organic amendments are a practical and sustainable strategy for boosting soil quality while mitigating erosive processes.

In a parallel effort, a separate study conducted in Ethiopia evaluated the influence of composted manure on surface runoff and erosion. The findings underscored composted manure's potential to diminish runoff volume and sediment loss, reinforcing its role as a soil-conserving amendment in erosion-prone landscapes [37]. The study found that the use of composted manure significantly reduced runoff and soil erosion. The application of composted manure increased the water infiltration rate, soil organic matter, and soil aggregate stability, which improved soil quality and reduced the potential soil erosion as shown in Table 3.

Table 3 Effect of intercropping practices on runoff and soil loss

Intercropping practices	Reduction in runoff (%)	Reduction in soil loss (%)	References
Gliricidia + Trench planting	29	45–48	[43]
Maize + cowpea	10.9–36	8.3–43	[44]
Maize + potato	15.65–80.20	33.96–85.70	[45–47]
Annual crop + strips of native perennial plants	34–89	45–94	[48]
Cereals + legumes	–	~46	[49]
Cassava + cowpea	–	40.0–58.20	[50]
Wheat + potato	–	50–93	[51]

In another study, evaluated the effect of organic inputs and inorganic fertilizers on runoff and soil loss in a semi-arid region of India [38]. The study found that the application of organic inputs significantly reduced runoff and soil loss compared to inorganic fertilizers. The use of organic inputs increased soil organic matter, water holding capacity, and soil aggregation, which resulted in improved soil structure and reduced soil erosion. The study concluded that organic fertilizers are an effective and sustainable way to improve soil quality and reduce soil erosion in semi-arid regions. The application of fresh pig manure proved instrumental in curbing soil erosion, primarily due to its rich organic matter content. This organic richness significantly boosted soil aggregation, fortifying the soil's structural framework. As a result, the enhanced cohesion among soil particles increased the soil's resilience against erosive forces, effectively reducing surface runoff and sediment loss [23]. Organic matter can also be combined with other erosion control practices to reduce soil loss in regions with intense agricultural activity [39]. Additionally, the use of organic matter, such as farmyard manure and poultry manure, has been found to reduce runoff and soil loss when combined with other techniques [40, 41]. Moreover, the use of contour tillage and straw mulching in combination with organic matter can effectively control soil erosion by reducing runoff depth and soil erodibility values [39].

Integrating manure application with seasonal fallowing has been shown to effectively curb soil erosion, as noted by [42]. This dual strategy enriches soil organic matter while allowing the land to rest, thereby enhancing soil structure and stability against erosive forces. The research suggests that the decomposition of manure can significantly increase soil organic matter, resulting in better soil structure and aggregation, which increases resistance to detachment. The use of organic matter and its decomposition in soil conservation practices has therefore been identified as an effective way to control soil erosion and improve soil quality. Overall, the use of organic inputs and manure decomposition can significantly reduce soil erosion by improving soil quality, structure, and moisture retention capacity. These practices can be particularly useful in regions where soil erosion is a significant issue, such as semi-arid regions in India and other parts of the world.

3.4 Tillage operation

Several studies were reviewed to investigate the effectiveness of tillage operations in controlling soil erosion and runoff in agricultural lands as shown in Table 4. The studies showed that various tillage methods, such as cross-ridge tillage, downslope tillage, and ridge tillage, combined with cover crops, mulching, or stubble retention, could significantly reduce soil erosion and runoff. No-tillage practices were also found to be effective in promoting soil health, reducing runoff, and improving infiltration. The reduction in soil erosion and runoff varied among different tillage methods and soil conditions. The study revealed that zero tillage significantly curbed surface runoff, reducing it by a factor of 1.51 to 2.53 when compared to traditional tillage methods [6]. The study highlighted that strategic crop combinations effectively minimized soil erosion: cocoa fields experienced the lowest erosion at 8.20 t ha^{-1} when ridges were paired with groundnut as a cover crop; areca plantations recorded just 4.64 t ha^{-1} with ridges intercropped with maize; and in oil palm systems, the ridge maize setup limited erosion to 12.33 t ha^{-1} [52]. Under intense rainfall conditions, runoff was curtailed by 10% with downslope tillage, while contour ridge tillage proved far more effective, slashing runoff by an impressive 49% [22]. Studies have shown that tillage operations, such as chiseling, subsoiling, and deep tillage, can reduce surface runoff and soil erosion in Indian agricultural lands [53–55]. Additionally, conservation tillage practices, such as zero tillage and reduced tillage, have been found to be effective in controlling soil erosion and improving soil health in Indian agricultural lands [56, 57]. However, the effectiveness of tillage as an erosion control technique may vary depending on factors such as soil type, slope, and weather conditions [58, 59].

Research in a rain-fed agricultural region of Indian Himalaya found that no-till practices combined with stubble retention led to a significant increase in cumulative infiltration and decrease in runoff. This resulted in a reduction of soil erosion by 48–50% [7]. In another study, [13] looked at the relationship between soil erosion and three tillage practices in a subtropical monsoon climate in a hilly soil area of Himalaya. The practices were minimum tillage, conventional tillage, and seasonal no-tillage, and the study found that seasonal no-tillage ridges reduced soil erosion and surface runoff compared to the other two methods. No-till practices can also increase the percentage of micro-pores in soil, which reduces runoff and improves infiltration. A study conducted in India revealed that adopting minimum tillage, when integrated with strategic vegetative barriers and well-planned nutrient management, significantly reduced both soil erosion and surface runoff. This conservation approach not only reduced the physical degradation of soil but also promoted better moisture retention and nutrient preservation. The synergy of reduced soil disturbance, protective plant cover, and tailored fertilization proved to be a resilient combination for sustainable land stewardship under Indian agro-ecological conditions [41].

Table 4 Effect of different tillage practices on runoff and soil erosion/loss

Tillage practice	Soil type	Crop/cropping system	Reduction in runoff (%)	Reduction in soil loss (%)	References
Ridge tillage opposite to slope direction	Loam	Cassava	42.6	61.7	[60]
Contour ridge tillage + mulch (3 t ha^{-1})	Sandy clay loam	Maize	29.2–33.3	11.6	[61]
No tillage + mulch (3 t $ha^{-1)}$			70.4–78.0	22.2–41.1	
Contour ridge tillage + mulch (3–5 t ha^{-1})	Sandy loam	Maize	37.5–55.6	20.4–36.3	
No tillage + mulch (3 t $ha^{-1)}$			54.7–78.4	21.5	
Conservation tillage + agro-geo-textile	Typic Udorthents	Maize	24.0	92.5–94.4	[36]
Minimum tillage	Sandy loam	Maize	13.2	25.8	[27]
Mulch tillage + cover crops (winter)	Silt loam	Corn-wheat	25–55	70–88	[62]
No tillage + cover crops during winter			49–60	84–93	
Conservation tillage	sandy loess	Maize	88.0–91.7	94.5–96.1	[63]
Conservation tillage (Direct drilled and stubble retained)	Greyish brown clay loam	Wheat-lupin	90.9	98.0	[64]
Down slope cultivation with no tillage	Oxisol	Wheat-soybean	80	21	[65]
Conservation tillage (reduced and no tillage)	Loamy Cambisol	Maize-wheat	–	73.2	[66]
No tillage	Silty-loam	Wheat-soybean	> 50	50–95	[67]

3.5 Cover crop

Cover crops are effective in reducing runoff and soil loss in India and around the world. In the semi-arid landscapes of Israel, where soil degradation and water scarcity pose persistent challenges, the strategic use of diverse cover crops in potato cultivation has emerged as a transformative agro-ecological approach. Integrating species such as rye, vetch, and clover into rotational cycles not only curtails soil erosion by reinforcing topsoil structure but also significantly reduces surface runoff through improved water infiltration. These living mulches act as a natural barrier against weed proliferation, competing effectively for light and nutrients, thus minimizing reliance on herbicides. The multi-functional role of cover crops combining physical soil protection with biological weed suppression demonstrates a promising shift toward sustainable potato farming in Israel's vulnerable agroecosystems [68]. The study found that runoff and soil erosion were reduced by 60% and more than 95%, respectively, among the five cover crops, oats were the most efficient. In dryland agricultural systems cultivating crops like soybean, radish, potato, and cabbage, the adoption of split fertilization strategies in conjunction with winter cover cropping has proven to be a powerful soil conservation tool. By synchronizing nutrient release with plant demand, split fertilization not only enhances nutrient uptake efficiency but also controls nutrient losses that contribute to sediment runoff. When paired with winter cover crops, this practice forms a dynamic buffer root systems stabilize the soil matrix while canopy cover shields it from erosive forces. Compared to conventional local practices, this integrative approach has led to a dramatic reduction in sediment yield, underscoring the synergy between nutrient management and vegetative soil protection in bolstering the resilience of dryland farming landscapes [69]. In Indonesia's erosion-prone upland and plantation terrains, the integration of *Mucuna bracteata* as a cover crop has shown considerable promise in soil and water conservation. When interwoven with a combination system of upland rice co-cultivated with soybean and strategically implemented strip planting of *Mucuna bracteata* within oil palm plantations, this approach created a layered defense against erosive forces. The dense, spreading habit of *Mucuna* not only blanketed the soil, reducing direct rainfall impact, but also slowed surface flow, effectively curbing runoff. This synergistic planting method markedly diminished soil erosion, illustrating how thoughtful species integration can harmonize crop productivity with landscape preservation in tropical agro-ecosystems [70]. These results highlight the potential of cover crops as a sustainable and effective practice for reducing runoff and soil erosion, both in India and globally.

3.6 Mechanical method

Mechanical methods for soil and water conservation have been thoroughly investigated across various global regions as shown in Table 5, providing valuable insights into their effectiveness. In the Indian context, several studies have underscored the remarkable effectiveness of soil and water conservation measures such as contour bunds, graded bunds, and furrows, particularly on sloping terrains cultivated with maize. These interventions have proven instrumental in curbing surface runoff and minimizing soil erosion, thereby enhancing moisture retention and sustaining soil fertility. By intercepting the flow of water and encouraging its gradual infiltration, these structures not only mitigate land degradation but also contribute to improved crop performance. Their strategic implementation on inclined maize fields stands as a testament to the synergy between traditional land management wisdom and scientific validation in promoting sustainable agriculture [71, 72]. The implementation of these mechanical measures resulted in a 36% and 34% reduction in runoff and a 44.3% and 44.1% reduction in soil loss, respectively.

Table 5 Impact of mechanical soil conservation measures on runoff and soil loss/erosion control

Mechanical measures	Reduction in runoff (%)	Reduction in soil loss (%)	References
Mulching	18–90	65–90	[28]
Hedgerows in the contours	10.9–36	80–99	[28]
Reduced/minimum tillage	9	23	[28]
Minimum tillage with plant strips	36–39	36	[75]
Soil bunds and cover crops	–	43–68	[76]
Drainage terraces	–	3.23–75	[84]
Filter strips	–	3.2–68	[84]
Drainage terraces + filter strips		57	[84]

On the sloping farmlands of China, a study explored the synergistic effects of ridge-furrow and contour cultivation two time-tested soil conservation techniques when employed in unison. This experimental fusion aimed to tackle the dual challenges of water runoff and soil erosion that plague inclined terrains. The ridge-furrow system, known for enhancing moisture retention, was aligned with the natural contours of the slope, creating a rhythmic, terraced pattern that not only slowed surface flow but also directed water inward for deeper infiltration. The contour alignment acted as a stabilizing agent, anchoring the ridged formations and mitigating the gravitational pull that often strips these landscapes of their fertility. Together, these practices revealed a promising agronomic choreography harmonizing water use efficiency with sustainable soil stewardship on vulnerable hillsides [73]. Their findings demonstrated a substantial decrease in surface runoff by 44.5% and soil loss by 50.3% compared to traditional cultivation methods. The success of these mechanical measures in reducing erosion underscores their potential for sustainable soil and water conservation.

In the rugged uplands of Nepal, where agriculture dances delicately with gravity, a study delved into the collective power of contour tillage, terracing, and vegetative barriers to tame the erosive force of sloped terrain. These mechanical interventions, each a sculptor of the land in its own right, were found to drastically transform the hydrological and erosional dynamics of the fields. The trio operated in concert terracing broke the descent into manageable steps, contour tillage aligned cultivation with the land's natural curves, and vegetative barriers stood as living sentinels, trapping sediment and calming runoff. The results were compelling: a 46% drop in runoff coefficient and a striking 52% decline in soil erosion rate compared to untreated control plots. This outcome underscored not just the efficacy but the necessity of structural soil conservation strategies in safeguarding Nepal's sloping agricultural heritage [74].

A significant decrease in surface runoff and soil erosion through the use of level terraces with bunds and horizontal terraces with Bahia grass (*Paspalum notatum*) [39]. In the United States, contour farming to be highly effective on sloping cropland, reducing surface runoff by 51% and soil loss by 39% compared to traditional methods [77]. Additionally, contour farming led to improvements in soil organic matter content and fertility. Broadening the conservation lens, a study in China turned its focus to the strategic deployment of check dams across hilly landscapes, aiming to intercept the restless energy of surface runoff before it could erode the terrain. These small but mighty structures, carefully positioned along the natural flow paths, acted as hydraulic buffers slowing down water velocity, encouraging sediment deposition, and stabilizing the surrounding soil. The impact was both immediate and impressive: surface runoff was curtailed by 40%, while soil erosion saw a substantial 38% decline [78]. This research highlighted the quiet resilience of check dams as vital guardians of sloped ecosystems, effectively bridging traditional engineering with sustainable watershed management. In Ethiopia, the positive impact of stone bunds, leading to a 38% reduction in surface runoff and a 32% decrease in soil loss [79]. In the undulating rice-growing regions of Bangladesh, a study zeroed in on contour trenching as a tactical intervention to counteract the erosive pressures of sloped cultivation. By carving shallow, strategically aligned trenches along the natural contours, the practice disrupted the downhill momentum of water, allowing more time for infiltration and minimizing soil displacement. The outcome was striking surface runoff dropped by 50%, and soil loss diminished by 45%, demonstrating the method's efficacy in nurturing both water retention and soil integrity. Especially in rice-farming systems, where standing water is essential, this approach proved to be a quiet revolution in managing hillside hydrology and preserving productivity on fragile terrain [80]. Amid Brazil's rolling agricultural landscapes, the introduction of grassed waterways emerged as a nature-based solution to the twin challenges of runoff and soil degradation. These vegetated channels, woven into the terrain like green arteries, guided excess water gently through the fields, curbing its erosive force while anchoring the soil with a living mesh of roots. The results spoke volumes: surface runoff was reduced by 48%, and soil loss fell by 40%, underscoring not only their effectiveness in erosion control but also their role in fostering biodiversity corridors. Beyond their utilitarian function, these verdant strips breathed ecological resilience into the farmland, blending agronomic efficiency with environmental harmony [81]. The aim of this study is to review the literature on soil erosion control practices in world agricultural land. With heavy precipitation, intensive farming, and varied topography, controlling soil erosion is crucial in various agricultural land. A thorough search of two databases yielded 55 relevant articles, identifying two main themes and several sub-themes. Between 2015 and 2023, research in many parts of the world primarily focused on biological methods, with tillage operations and mulching being the most commonly used and recommended practices. Due to a diverse range of climate regimes, including tropical, subtropical, Mediterranean, and temperate, and the discussion of results focuses on specific countries in light of these climates [73].

Soil erosion poses a significant environmental challenge, resulting in loss of top soil, compromised water quality, and reduced agricultural productivity [7]. In India, the annual loss of about one billion metric tons of topsoil is a pressing concern, prompting the implementation of diverse soil erosion control practices [16]. These practices encompass tillage methods, mulching, cover crops, grass cultivation, and mechanical techniques like terracing and contour farming. While tillage practices like no-tillage and conservation tillage offer proven benefits for soil structure, their adaptability varies

based on local climate and ecosystem conditions [82]. Mulching, a widely employed erosion control method, comes in organic and inorganic types, selected based on land use and climate. Organic mulch, gradually decomposing, enriches the soil, retains moisture, and improves structure [54]. Despite being effective in erosion control, cover crops cultivation may present a challenge by competing with main crops for soil water, potentially leading to reduced yields [33]. Mechanical methods, including terracing to intercept water flow, contribute to conserving soil moisture and controlling erosion. In India's tropical climate, conventional tillage practices are common, but recent studies suggest that adopting minimum tillage, combined with strategies like weed mulching, vegetative barriers, and organic matter application, proves more effective [2]. The incorporation of organic matter, such as manure, enhances enzyme activity and microbial biomass, mitigating splash erosion and delaying runoff [70].

Advocating for sustainable yield boosts, the intercropping of cover crops like sweet potato, turmeric with major crops and selective weed retention may be adopted [15]. This comprehensive approach underscores the need for multifaceted strategies to address soil erosion challenges in India's agricultural landscape [17]. Global agriculture grapples with soil erosion, prompting diverse strategies for control. In the U.S., practices like conservation tillage are used [83], while China employs terraces and conservation tillage [39]. Worldwide, organic matter application proves effective in enhancing soil and reducing erosion [42]. The success of no-tillage practices relies on local climate and ecosystem conditions, demanding customized implementation [17].

The Food and Agriculture Organization highlights its substantial benefits for climate adaptation, enhancing crop resilience in challenging conditions. No-tillage, as evidenced in systematic reviews, encourages water retention, stable surface aggregates, and increased total porosity [54]. Globally, various soil erosion control practices like tillage, mulching, cover crops, grass cultivation, and mechanical methods are employed, each with its pros and cons [22, 53, 82]. The efficacy depends on specific soil and ecosystem conditions, with cost as a potential limitation for practices like cover crops [55, 56]. Mechanical methods, less studied but including basin leaching and contour trenching, are a last resort when biological methods fall short. While biological methods are preferred for their lower cost and minimal landscape impact, the article suggests that a combination of biological and mechanical methods proves more effective for controlling soil erosion in agricultural land.

3.7 Impacts of biological and mechanical measures on soil properties

Integrating biological and mechanical soil conservation measures significantly enhances soil health and productivity across diverse agro-climatic zones in India. Conservation agriculture practices in the Indo-Gangetic Plains, such as zero tillage and residue retention, have led to a 43% increase in soil organic carbon and a 56% rise in microbial biomass carbon compared to conventional tillage systems [93]. In the laterite soils of India's west coast, long-term implementation of conservation measures improved soil structure and fertility [94]. In Meghalaya, bench terracing significantly reduced erosion and improved nutrient availability, contributing to higher crop yields [95]. Similarly, the Vetiver System, using vetiver grass hedgerows, has proven effective for slope stabilization and erosion control due to its dense, deep root system [96]. Khokhar et al. demonstrated that maize–cowpea intercropping in the Shivalik foothills reduced runoff by 10.9% and improved maize-equivalent yields [97]. Modak et al. found that zero tillage improved soil aggregation and carbon stabilization in subtropical India [98].

4 Conclusion and future direction

Soil erosion is a significant problem in agricultural land, and different soil conservation practices have been implemented to reduce soil loss. Research on control practices for soil erosion in agricultural land has focused more on biological methods than mechanical methods. However, future studies should focus on mechanical methods for soil erosion control for longer time periods, to determine their effectiveness in controlling soil erosion. One area for future research is the use of cover crops to reduce soil erosion, which has been shown to be effective in reducing runoff and erosion rates. The best cover crops for different soil and climatic conditions need to be determined, as well as the optimal timing and methods for planting and managing them. The effectiveness of different soil conservation practices under changing climate conditions also needs to be studied to reduce the impacts of soil erosion on crop yields and the environment. The review highlights the importance of tillage operation and mulching as the most effective and preferred methods to be practiced in agricultural land. Proposed subsidy structures for smallholders adopting conservation tillage at initial stage.

It also stresses the importance of support from local government regarding financial management, training, and policies to control soil erosion. Combining both biological and mechanical methods could be more effective in controlling soil erosion in agricultural land. Soil erosion is a significant challenge in agricultural land, and there is still much work to be done to determine the most effective and practical methods for different agricultural landscapes. Research should focus on cover crops, the effectiveness of mechanical methods, and the adaptation of soil conservation practices to changing climate conditions. By continuing to study and refine these methods, we can work towards a more sustainable and resilient agricultural system that is better able to withstand the impacts of soil erosion.

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Declarations

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